

1. Dari survey penghasilan per-kepala per-hari orang Indonesia, diambil dari 81 orang sebagai Sampel, menghasilkan rata-rata sampel Rp. 125.000,- dan simpangan baku sampel Rp 30.000,-. Hitunglah confidence interval μ terhadap rata-rata sampel, jika confidence levelnya:

a) 95%

$$\alpha = 0,05 \quad n = 81 \text{ orang} \quad S = \text{Rp } 30.000$$

$$\frac{1}{2}\alpha = 0,025 \quad \bar{x} = \text{Rp } 125.000$$

$$Z_{0,025} = 1,96$$

$$\left[125.000 - 1,96 \frac{30.000}{\sqrt{81}} ; 125.000 + 1,96 \frac{30.000}{\sqrt{81}} \right]$$

$$[118.466,67 ; 131.533,33]$$

b) 98%

$$\alpha = 0,02 \quad Z_{0,01} = 2,33$$

$$\frac{1}{2}\alpha = 0,01$$

$$\left[125.000 - 2,33 \frac{30.000}{\sqrt{81}} ; 125.000 + 2,33 \frac{30.000}{\sqrt{81}} \right]$$

$$[117.233,33 ; 132.766,67]$$

c) 99%

$$\alpha = 0,01 \quad Z_{0,005} = 2,58$$

$$\frac{1}{2}\alpha = 0,005$$

$$\left[125.000 - 2,58 \frac{30.000}{\sqrt{81}} ; 125.000 + 2,58 \frac{30.000}{\sqrt{81}} \right]$$

$$[116.400 ; 133.600]$$

d) Six-sigma

$$Z_{\frac{1}{2}\alpha} = 3$$

$$\left[125.000 - 3 \frac{30.000}{\sqrt{81}} ; 125.000 + 3 \frac{30.000}{\sqrt{81}} \right]$$

$$[115.000 ; 135.000]$$

2. Seperti soal nomor 1, dengan asumsi populasi berdistribusi normal, tetapi hanya disurvei 9 orang saja, hasil survey (dalam ribuan Rp):
100, 2000, 30, 5000, 4000, 200, 500, 250, 1000

$$\bar{x} = \text{Rp } 1.453.333,33$$

$$s^2 = 3.419.225,000$$

$$s = 1.849.114,653$$

a) 95%

$$t_{0,025,8} = 2,306$$

$$\left(1.453.333 \pm 2,306 \frac{1.849.114,653}{\sqrt{9}} \right)$$

$$(31.980,7 ; 2.874.685,3)$$

b) 98%

$$t_{0,01,8} = 2,896459$$

$$\left(1.453.333 \pm 2,896459 \frac{1.849.114,653}{\sqrt{9}} \right)$$

$$(-331.961,93 ; 3.238.627,93)$$

c) 99%

$$t_{0,005,8} = 3,355387$$

$$\left(1.453.333 \pm 3,355387 \frac{1.849.114,653}{\sqrt{9}} \right)$$

$$(-614.832,09 ; 3.521.498,09)$$

d) Six-sigma

$$t_{\frac{\alpha}{2}, n-1} = 3$$

$$\left(1.453.333 \pm 3 \frac{1.849.114,653}{\sqrt{9}} \right)$$

$$(-395.781,65 ; 3.302.447,65)$$